

V-shaped trajectory planning of search and rescue UAV based on target detection

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Abstract—In emergency situations characterized by frequent natural disasters and maritime fatalities, unmanned aerial vehicles (UAVs) are increasingly employed in search and rescue missions due to their superior line-of-sight capabilities and operational flexibility. However, factors such as the small size of UAV platforms, high-altitude operation and fast flight speed make the implementation of search and rescue missions often face the uncertainty of target detection. To address this challenge, this paper proposes a straightforward and effective V-shaped trajectory planning method. When the UAV captures a potential target in the air, it will fly diagonally down to get a clearer image of the target, then diagonally back up after accurate identification. The strategy significantly enhances target recognition accuracy by optimizing the UAV trajectory. Simulation results indicate that, compared to traditional straight flight trajectory, the V-shaped trajectory not only ensures higher recognition accuracy but also markedly improves stability in target detection. Notably, in the case of high recognition accuracy, this method can still complete the missions well. Consequently, this method offers a novel solution for enhancing the efficiency of UAV-based search and rescue operations.

Keywords—target detection, UAV search and rescue, V-shaped trajectory

I. INTRODUCTION

The increasing frequency of natural disasters worldwide, along with emergencies such as maritime accidents, underscores the critical importance of rapid search and rescue operations for missing persons. In this context, unmanned aerial vehicles (UAVs) have gained widespread application in search and rescue missions due to their exceptional line-of-sight capabilities and high mobility [1]. UAVs are capable of carrying a variety of sensors and global navigation satellite systems (GNSS), enabling autonomous flight and obstacle avoidance functions. Additionally, they can identify and locate ground targets using onboard cameras. The overall UAV rescue system encompasses functions such as target detection, target positioning, and autonomous trajectory planning. Therefore, optimizing these functions collectively is essential for enhancing the efficiency of UAV-based rescue operations.

In the realm of object detection within aerial imagery, Zhong et al. [2] utilized images captured by visible light cameras to detect and recognize human body areas based on an improved depth-separable multi-head network structure employing YOLO v8. However, under dim conditions, visible light images often suffer from loss of contrast and

detail. To address this issue, Li et al. [3] proposed a deep learning-based method for detecting targets in thermal images captured by UAVs that operates effectively in both day and night scenarios. Zhao et al. [4] employed the same convolution operator for feature extraction from both visible light and thermal images while successfully mitigating the loss of important features through image reconstruction techniques. Furthermore, Torabi et al. [5] introduced a video dataset comprising both visible light and thermal imaging data; this dataset was collected at a rate of 30 frames per second under various tracking conditions.

After the UAV identifies a potential target, it is essential to confirm the relative coordinates or geographic coordinate information of that target. Ahmed investigated the impact of varying flight altitudes on the geometric accuracy of generated topographic map products, examining spatial resolution across seven different flight heights and conducting multivariate nonlinear regression analysis. The findings indicated accuracies of 92.76% and 91.91%, respectively [6]. Minaeian et al. [7] employed a monocular camera integrated with image processing technology on the UAV to facilitate target localization within RGB space.

In complex disaster relief operations, unmanned aerial vehicles (UAVs) possess the capability to autonomously assign tasks or make decisions [8]. Zheng et al. [9] modeled the UAV search area as a small square grid consisting of L_x rows and L_y columns, treating each grid cell as an independent search zone. Ruan et al. [10] assumed that the UAV operated at a fixed altitude and established its coordinates on a two-dimensional plane where these coordinates were linked to speed and steering angle. In light of GPS positioning uncertainties, Yin et al. [11] utilized a simplified binary Gaussian model to approximate this uncertainty while providing independent and isotropic horizontal accuracy for GPS measurements. Masadeh et al. [11] developed an autonomous UAV system designed for dynamic intrusion detection, addressing challenges related to energy constraints and target motion unpredictability. Michael et al. [12] proposed a UAV-based model for target detection and tracking that leverages long short-term memory networks (LSTM) for effective monitoring purposes. Yang et al. [13] introduced a layered distributed framework aimed at implementing deep learning-based target tracking solutions.

Numerous alternatives are offered by the aforementioned literature, including boosting the target tracking efficiency, target locating capability, and picture recognition method.

However, the UAV may experience a lower target recognition accuracy in the real application environment, leading to target omission or placement inaccuracy, as a result of bad shooting angles or object occlusion. Owing to the significance of optimizing the UAV's flight trajectory while carrying out its mission, the possibility of enhancing target detection precision through flight trajectory modifications has not been thoroughly investigated in the current research.

Earlier work by this research team [14] proposed that a V-shaped trajectory can reduce mission execution time while guaranteeing the integrity of information collecting. Following this, more study [15] proved the effectiveness of the inverted trapezoid trajectory for inclined downward flight when performing air missions with known start, target, and end sites. Inspired by these findings, this research suggests a novel V-shaped trajectory planning method for boosting target detection accuracy by optimizing the UAV's flying route. This method is based on the relative position of the UAV and the ground target. When the UAV finds a potential target for the first time, it will change its flight trajectory, tilt down to approach the target for identification. After confirming the target, the UAV oblique upward to return to the cruise path. The whole target detection trajectory is V-shaped. This method can produce crisper image, increasing the accuracy of target detection. In addition, if the target cannot be validated at the lowest flight height, the UAV sends an image of the potential target to the data center for verification by an operator. The simulation analysis demonstrates that, when compared to the traditional straight flight trajectory, the V-shaped trajectory method not only significantly improves target detection stability while ensuring recognition accuracy, but it can also effectively shorten search and rescue time under various recognition accuracy thresholds. This strategy offers a fresh perspective for increasing the efficiency of UAVs in search and rescue operations.

The rest of this article is structured as follows. The section II introduces the system model and problem formulation, describing the flight trajectory and mathematical model within the UAV search and rescue range. Section III introduces the V-shaped trajectory of UAV target detection. Section IV introduces the evaluation and analysis of the proposed scheme. Finally, in the fifth section, the proposal is summarized.

II. SYSTEM MODELING AND PROBLEM PRESENTATION

A. System Model

The UAV equipped with image recognition is responsible for search and rescue of targets on the ground within a rectangular region of $N_{km} \times N_{km}$. Assume that the target's exact location is unknown, and that there are N potential targets randomly distributed around the area, of which only one is correct. Assumed the target stationary, the search and rescue operation must cover the entire area. In this article, the UAV employs a Snake Scan mode to provide the most comprehensive survey of the entire area. During routine detection, the UAV's flight altitude is set to H , speed to V , and battery capacity to E . The UAV's camera module is capable of taking images of the ground in the $m \times m$ range from directly below it to in front of it. When the UAV runs low on energy, a return mission will be performed to replace

the battery and continue the detection, as shown in Fig. 1.

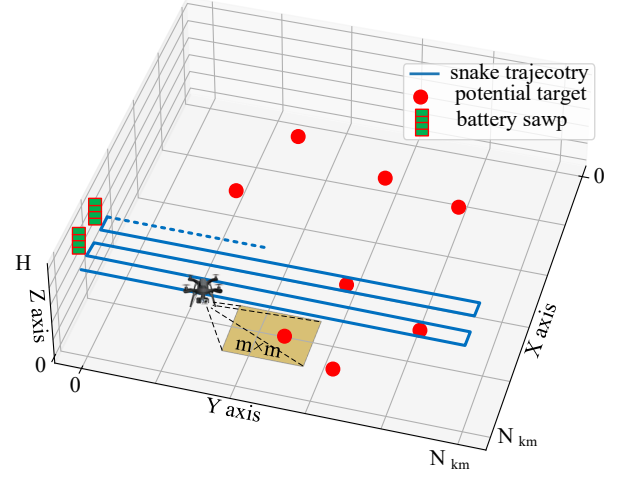


Fig. 1. UAV Search and Rescue system model diagram.

B. Problem presentation

In the process of aerial detection of ground targets, the shooting position of UAV is poor, the target is blocked and so on, which leads to low recognition accuracy and missing detection. Suppose that the probability of the UAV's target detection system identifying a potential target as a rescue target is

$$P_R(x, y, z) = \begin{cases} 1 & (x, y, z) \rightarrow 0 \\ 0 & \end{cases} \quad (1)$$

Where (x, y, z) refers to the relative 3D coordinates between the UAV and the ground target. Setting the closer the UAV is to the target, the higher the recognition probability. That is, the probability of the rescue target approaches 1, or the probability of the rescue target approaches 0. This paper assumes that there is no fraud. There is no false target acting as the rescue target.

During shooting, the UAV camera module is usually affected by shaking, light, weather conditions and focusing, etc., resulting in certain identification errors, denoting γ . In addition, the accuracy of the target detection model itself will also produce some error in the target detection, which is denoted as ρ . Therefore, the probability that the UAV recognizes the potential target as the rescue target at a certain position is rewritten as

$$P_T(x, y, z) = P_R(x, y, z)\rho\gamma \quad (2)$$

Then the probability that it is not a rescue target is $P_F(x, y, z) = 1 - P_T(x, y, z)$. In the search and rescue process, any potential target may be the rescue target to be sought. Therefore, the UAV should focus on the detection of the potential target during flight. When the probability of identifying it as a rescue target is small, the probability that it is not a rescue target is high. Combining the above two formulas, the recognition accuracy rate of the target is set as

$$P(x, y, z) = |P_T(x, y, z) - 0.5| + 0.5 \quad (3)$$

Where $|\cdot|$ is the absolute value. It can be concluded

from the above formula that the target recognition accuracy of the system is related to the relative positions of the UAV and the potential target. And the final recognition accuracy of each potential target should be satisfied.

$$P(x, y, z) \geq \varepsilon \quad (4)$$

Where, ε is the threshold value. When the recognition accuracy of the target is less than ε , it is called a potential target and needs to be further determined.

In search and rescue missions, the goal is to minimize the search time in the area while ensuring that the accuracy of each target is within the threshold. The whole process consists of three parts: 1) normal patrol time; 2) target confirmation time when a potential target is found; And 3) the time it takes for the UAV to replace its battery, as shown in Figure 1. In the normal patrol phase, the UAV flies according to the predetermined snake scanning trajectory and recognizes the collected ground images; In the second stage, when the UAV finds a potential target, it needs to change the trajectory to further determine whether the target is a rescue target. In the third stage, the UAV changes its batteries by itself after completing a U-shaped trajectory. The UAV's search and rescue time can be calculated by the following formula:

$$\min_{(x,y,z)} T = \sum T_n + \sum_{i=1}^N T_{c,i} + \sum T_e \quad (5)$$

$$s.t \quad P(x, y, z) \geq \varepsilon$$

$$\sum T_n E_n + \sum_{i=1}^N T_{c,i} E_c \leq E$$

Where, T_n is a time of the UAV's normal patrol, $T_{c,i}$ is the target confirmation time of the potential target i , T_e is the UAV's battery change time, E_n is the energy consumption of the normal patrol, and E_c is the energy consumption of the potential target confirmation. In the above formula, $\sum T_n$ is also the flight time of the UAV along a fixed snake trajectory, with cannot be further optimized. The position of the changing station is fixed, and the UAV carries out power change once each back and forth, and $\sum T_e$ has no optimization space. Therefore, the confirmation time of the potential target of the UAV can be optimized, and the trajectory of the UAV can be optimized to reduce T_c under the condition of ensuring the target recognition accuracy and the smooth return of the UAV's remaining energy.

To sum up, the goal is to optimize the UAV's recognition trajectory and minimize the UAV's search and rescue time in the whole area under the condition of meeting the recognition accuracy and energy limitation.

III. V-SHAPED SEARCH TRAJECTORY METHOD

Assuming that the target is randomly distributed in the range, the snake trajectory is used for traversal. The solution of formula (5) is mainly affected by the constraints of formula (4). When the UAV detects a potential target for the first time, in order to further confirm the target, the position

of the UAV relative to the potential target should be changed so that the camera module can take clearer images. This paper assumes that the closer the UAV is to the target in the same viewing Angle, the higher the recognition accuracy. Previous studies [14] have demonstrated that adopting a V-shaped trajectory during the communication between the UAV and the target can complete missions in the shortest possible time. Inspired by this, this paper proposes a V-shaped search and rescue trajectory, as shown in Fig. 2.

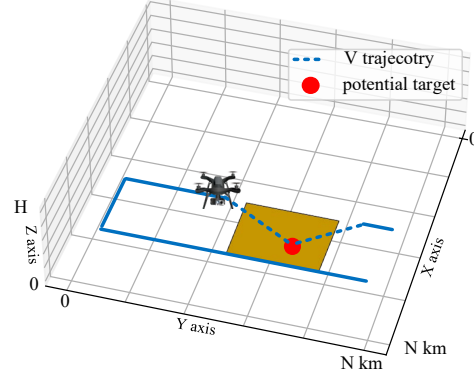


Fig. 2. V-shaped trajectory of UAV search and rescue.

The time taken by the UAV to identify a potential target using the V-shaped trajectory can be expressed as the sum of the time of oblique downward flight and oblique upward flight

$$T_{c,i} = t_{down} + t_{up} \quad (6)$$

Where, t_{down} and t_{up} represent the time of oblique downward flight and oblique upward flight, respectively.

When the UAV finds the potential target, it obtains the coordinate information of the potential target through the image positioning technology [6]. And immediately changes the flight trajectory, and flies downward toward the potential target on the ground. After confirming the target, the UAV returns to its normal trajectory oblique upward. If the target cannot be confirmed at the lowest altitude, the UAV uploads an image of the potential target to the data center for verification by staff. The overall process is as follows:

1. The UAV patrols the ground targets follow a snake trajectory.
2. When potential targets are found, i.e. $P(x, y, z) < \varepsilon$. Calculate the coordinates of potential targets based on existing image positioning techniques.
3. **while** ($|UAV - GN| > r$ and $P(x, y, z) < \varepsilon$)
 The UAV flies oblique downward while continuously recognizing potential targets.
end while
5. **if** The accuracy rate of recognizing the target is satisfied $P(x, y, z) \geq \varepsilon$:
if The target is a rescue target:
 Sends commands to the data center and tracks the target.
else if The target is not a rescue target:
 Then fly oblique upward to return to the predetermined trajectory to continue search and rescue.
end if
- else:**
 The system can't accurately recognize the target, upload the image of the potential target for the staff to process, then oblique upward flight back to the predetermined trajectory to continue the search and rescue.
end if

IV. SIMULATION RESULTS AND DISCUSSION

In this section, the performance of the proposed method is verified. Without loss of generality, the UAV flies at a speed of 10m/s, an altitude of 50m, and a search range of 1km*1km. The UAV changes a battery every round trip, and the battery change time is 30s. During the search and rescue process, there are 20 potential targets randomly distributed within the search range, and the target being searched is one of them. Set the accuracy rate of the target detection model itself $\rho = 0.98$, the probability $P_R(x, y, z)$ of all potential targets in (0~0.95) random generation, the error disturbance γ in the recognition process is a normal distribution with an average of 1 and variance of 0.05, and the threshold of the model's final recognition accuracy rate is $\varepsilon = 0.95$. It is assumed that the smaller the relative distance between the UAV and the ground target is, the larger the P_R is.

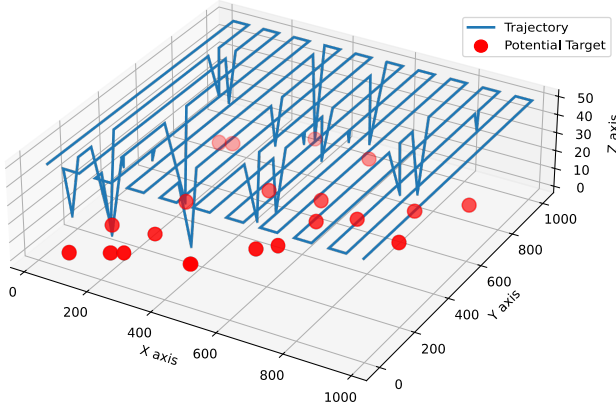


Fig. 3. UAV search and rescue trajectory planning for the proposed V-shaped trajectory.

The Fig. 3 shows the V-shaped trajectory proposed by this method. It can be seen from the Fig. 3 that the UAV flies towards the predetermined serpentine trajectory. Whenever the detection system finds a potential target, the UAV flies diagonally downward towards the target and approaches the potential target to improve the recognition accuracy of the target. When the accuracy reaches the setting requirements, the UAV returns to the normal cruise trajectory diagonally upward. In addition, when the potential target is clear, the UAV has a smaller V-shaped trajectory; when the target is unclear, the UAV needs to approach the recognition. Therefore, for outdoor search and rescue with high uncertainty factors, this method can obtain higher recognition accuracy.

In this part, we verify the recognition accuracy of the proposed scheme and compare it with the traditional serpentine trajectory direct flight method. In order to simulate the target recognition accuracy in the direct flight mode, this paper adopts a normal distribution with the initial potential target probability as the average, and the final recognition rate of the direct flight mode is the maximum value in the detection area. As shown in the Figure 4, the recognition accuracy rate of V-shaped trajectory is stable because it is recognized when approaching the target. In contrast, the direct flight trajectory can only slightly improve the recognition accuracy by fine-tuning the flight Angle, but the problem of unclear image due to altitude still exists. Therefore, this method realizes short-range and multi-angle recognition through V-shaped trajectory, which can greatly improve the recognition rate of potential targets, thus

enhancing the credibility of judging them as rescue targets.

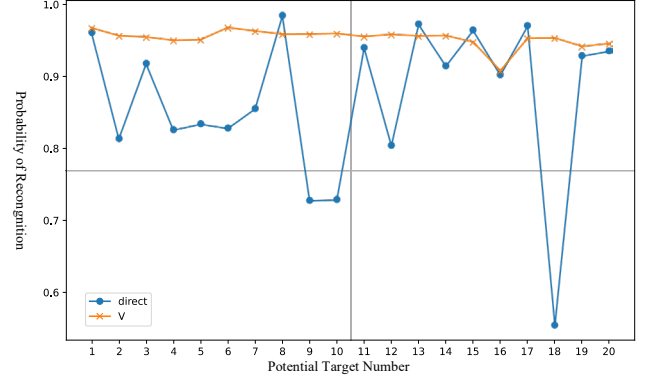


Fig. 4. Comparison of recognition accuracy rates of potential targets.

In this part, we verify the effect of recognition threshold on search and rescue time. As shown in Table I, the direct flight trajectory in the comparison method adopts a fixed trajectory, resulting in a fixed search and rescue time of 2400 seconds for the whole area. As can be seen from the table, the higher the threshold ε of the recognition system, the higher the accuracy of the recognition of each potential target, but the search and rescue time required also increases. However, due to the fixed direct flight trajectory, the system cannot ensure that each potential target can achieve a high recognition accuracy. Therefore, the proposed method provides an effective trajectory planning method for accurate search and rescue.

TABLE I. INFLUENCE OF IDENTIFICATION THRESHOLD ON SEARCH AND RESCUE TIME.

ε	0.6	0.7	0.8	0.9	0.95
V-shape method missions time / s	2448	2505	2548	2678	2782

V. CONCLUSION

This paper introduces a UAV search and rescue trajectory planning method based on V-shaped trajectory, which improves the accuracy of identifying potential targets on the ground by optimizing the flight trajectory of the UAV. In the simulation environment, the proposed V-shaped trajectory is compared with the traditional straight-line flight mode. The results show that the V-shaped trajectory can provide a clearer view when approaching the target, thus significantly improving the stability and accuracy of recognition. In addition, through the analysis of the search and rescue time under different recognition accuracy thresholds, it is confirmed that the proposed method can still complete the search work in the whole search and rescue area in less time under high recognition accuracy requirements. In summary, this method provides a feasible solution for improving UAV search and rescue efficiency.

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